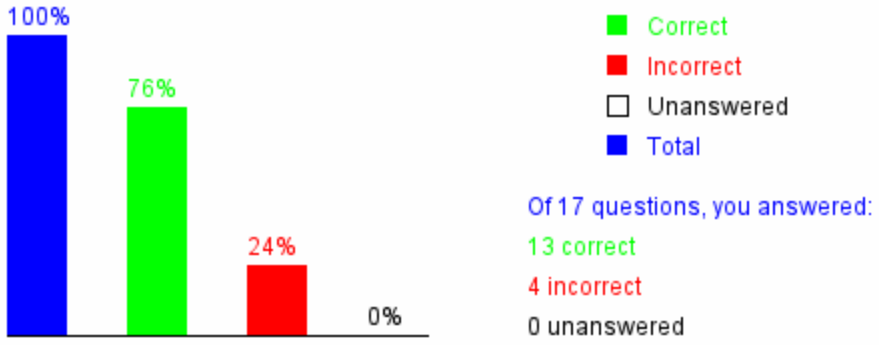
Introduction to Programming Using Python, Y. Daniel Liang

Many questions in this edition have been updated in the new edition. Please check with the publisher on the newest edition.

This quiz is for students to practice. A large number of additional quiz is available for instructors using Quiz Generator from the Instructor's Resource Website. Videos for Java, Python, and C++ can be found at https://yongdanielliang.github.io/revelvideos.html.

Chapter 7 Objects and Classes



Please send suggestions and errata to Dr. Liang at y.daniel.liang@gmail.com. Indicate which book and edition you are using. Thanks!

Section 7.2 Defining Classes for Objects

7.1 _____ represents an entity in the real world that can be distinctly identified.

- A. `A class`
- B. `An object`
- C. `A method`
- D. `A data field`

Your answer is correct

7.2 _____ is a template, blueprint, or contract that defines objects of the same type.

- A. `A class`
- B. `An object`
- C. `A method`
- D. `A data field`

Your answer is correct

7.3 An object is an instance of a _____.

- A. `program`
- B. `class`
- C. `method`
- D. `data`

Your answer is correct

7.4 The keyword _____ is required to define a class.

- A. `def`
- B. `return`
- C. `class`

D. All of the above.

Your answer is correct

7.5 _____ is used to create an object.

- A. A constructor
- B. A class
- C. A value-returning method
- D. A None method

Your answer is correct

7.6 The _____ creates an object in the memory and invokes _____.

- A. the __init__ method
- B. the init method
- C. the initialize method
- D. the __str__ method

Your answer is correct

7.7 Analyze the following code:

```
class A:
    def __init__(self, s):
        self.s = s

    def print(self):
        print(s)

a = A("Welcome")
a.print()
```

- A. The program has an error because class A does not have a constructor.
- B. The program has an error because class A should have a print method with signature print(self, s).
- C. The program has an error because class A should have a print method with signature print(s).
- D. The program would run if you change print(s) to print(self.s).

Your answer is correct

7.8 Analyze the following code:

```
class A:
    def __init__(self, s):
        self.s = s

    def print(self):
        print(self.s)

a = A()
a.print()
```

- A. The program has an error because class A does not have a constructor.
- B. The program has an error because s is not defined in print(s).
- C. The program runs fine and prints nothing.

D. The program has an error because the constructor is invoked without an argument.

Your answer C is incorrect
The correct answer is D

7.9 Analyze the following code:

```
class A:
    def __init__(self, s = "Welcome"):
        self.s = s

    def print(self):
        print(self.s)

a = A()
a.print()
```

A. The program has an error because class A does not have a constructor.
B. The program has an error because s is not defined in print(s).
C. The program runs fine and prints nothing.
D. The program has an error because the constructor is invoked without an argument.
E. The program runs fine and prints Welcome.

Your answer is correct

7.10 Given the declaration x = Circle(), which of the following statement is most accurate.

A. x contains an int value.
B. x contains an object of the Circle type.
C. x contains a reference to a Circle object.
D. You can assign an int value to x.

Your answer B is incorrect
The correct answer is C

Section 7.4 Hiding Data Fields

7.11 Analyze the following code:

```
class A:
    def __init__(self):
        self.x = 1
        self.__y = 1

    def getY(self):
        return self.__y

a = A()
print(a.x)
```

A. The program has an error because x is private and cannot be access outside of the class.
B. The program has an error because y is private and cannot be access outside of the class.
C. The program has an error because you cannot name a variable using __y.
D. The program runs fine and prints 1.
E. The program runs fine and prints 0.

Your answer is correct

7.12 Analyze the following code:

```
class A:
    def __init__(self):
        self.x = 1
        self.__y = 1

    def getY(self):
        return self.__y

a = A()
print(a.__y)
```

- A. The program has an error because x is private and cannot be access outside of the class.
- B. The program has an error because y is private and cannot be access outside of the class.
- C. The program has an error because you cannot name a variable using __y.
- D. The program runs fine and prints 1.
- E. The program runs fine and prints 0.

Your answer is correct

7.13 Analyze the following code:

```
class A:
    def __init__(self):
        self.x = 1
        self.__y = 1

    def getY(self):
        return self.__y

a = A()
a.x = 45
print(a.x)
```

- A. The program has an error because x is private and cannot be access outside of the class.
- B. The program has an error because y is private and cannot be access outside of the class.
- C. The program has an error because you cannot name a variable using __y.
- D. The program runs fine and prints 1.
- E. The program runs fine and prints 45.

Your answer is correct

7.14 In the following code,

```
def A:
def __init__(self):
    __a = 1
    self.__b = 1
    self.__c__ = 1
    __d__ = 1

# Other methods omitted
```

Which of the following is a private data field?

- A. __a
- B. __b
- C. __c__
- D. __d__

Your answer is correct

7.15 Analyze the following code:

```
class A:
    def __init__(self):
        self.x = 1
        self.__y = 1

    def getY(self):
        return self.__y

a = A()
a.__y = 45
print(a.getX())
```

- A. The program has an error because x is private and cannot be access outside of the class.
- B. The program has an error because y is private and cannot be access outside of the class.
- C. The program has an error because you cannot name a variable using __y.
- D. The program runs fine and prints 1.
- E. The program runs fine and prints 45.

Your answer is correct

7.16 Which of the following statement is most accurate?

- [x] A. A reference variable is an object.
- [x] B. A reference variable refers to an object.
- [x] C. An object may contain other objects.
- [x] D. An object may contain the references of other objects.

Your answer ABCD is incorrect
Click here to show the correct answer

7.17 What is the value of times displayed?

```
def main():
    myCount = Count()
    times = 0

    for i in range(0, 100):
        increment(myCount, times)

    print("myCount.count =", myCount.count, "times =", times)

def increment(c, times):
    c.count += 1
    times += 1
```

```
class Count:
    def __init__(self):
        self.count = 0

main()
```

- A. `count is 101 times is 0`
- B. `count is 100 times is 0`
- C. `count is 100 times is 100`
- D. `count is 101 times is 101`

Your answer D is incorrect 
Click here to show the correct answer